

7 A Hall probe consists of a slice of conducting material, as shown in Figure 7.1.

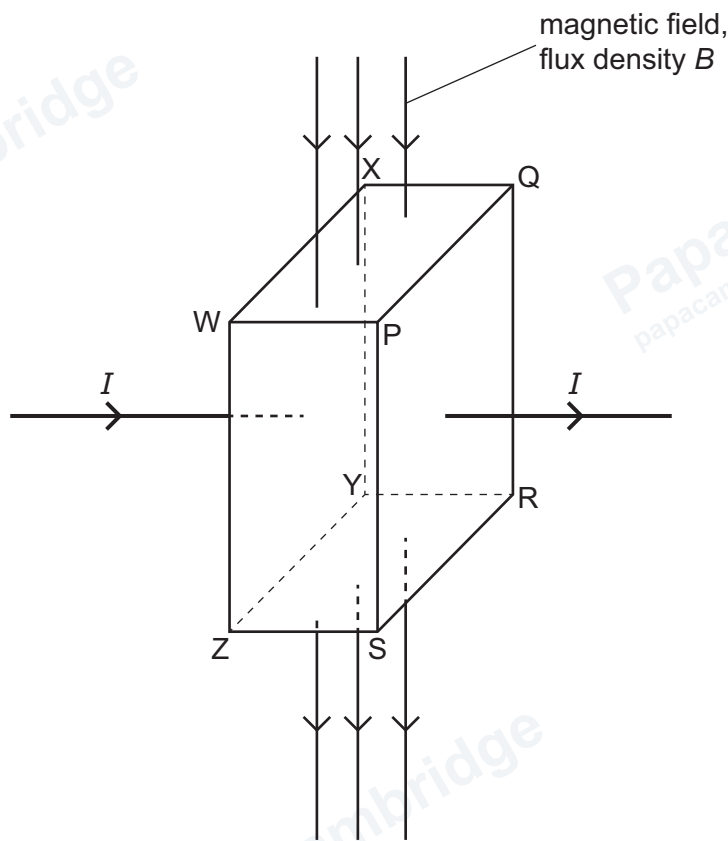


Figure 7.1

Current I enters the slice at face WXYZ and leaves it at face PQRS.

The slice is in a uniform magnetic field of flux density B . This magnetic field is perpendicular to faces WXQP and ZYRS.

A Hall voltage V_H is induced between two of the faces of the slice.

- (a) (i) Using the letters P, Q, R, S, W, X, Y and Z, identify the faces of the slice between which the Hall voltage is induced.

..... and [1]

(ii) The magnitude of the Hall voltage in Figure 7.1 is given by the equation

$$V_H = \frac{BI}{nqt}.$$

State the meanings of the symbols n , q and t in this equation.

n

q

t

[3]

(b) (i) Explain the origin of the Hall voltage.

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..... [3]

(ii) Identify which of the faces in 7(a)(i) is at the higher potential. Explain your reasoning.

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..... [1]

[Total: 8]